

StableSort-CMC: a tracking algorithm for robot dog based orchard fruit counting

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Abstract—In recent years, the utilization of deep learning technology for fruit yield prediction methods using robot dogs equipped with cameras, along with target detection and tracking algorithms, has emerged as a research focus. However, the movement of robot dogs through undulating orchard terrain often induces vibration in the mounted cameras, which can significantly impact the performance of tracking algorithms. To address this issue, this paper proposes a tracking algorithm integrated with a camera motion compensation module. This module applies image rotation, scaling, and translation parameters to the tracking algorithm, enhancing its performance and counting accuracy in vibrated scenes. Utilizing nectarine data collected by a robot dog equipped with a gimbal camera, the proposed algorithm achieves a Multiple Object Tracking Accuracy (MOTA) of 63.4%. In ablation experiments, integrating the camera motion compensation module into the tracking algorithm improves MOTA by 2.9%. These results validate the efficacy of the proposed algorithm in mitigating the effects of camera vibration and enhancing tracking performance in dynamic orchard environments

Keywords—fruit count, target tracking, robot dog, camera motion compensation

I. INTRODUCTION

With the modernization and intelligent development of agricultural production, orchard management has put forward higher requirements for the accuracy and efficiency of fruit counting. Yield statistics through fruit counting plays a crucial role in the development of fruit harvest operation plans and marketing strategies [1]. At the present stage, accurate estimate of orchard yields before harvesting fruits helps managers to rationally allocate labor according to the distribution of fruits in the park [2]; in the future, it helps to deploy harvesting robots working strategies to efficiently complete the harvesting tasks and realize intelligent production in the orchard. Traditionally, the prediction method using manual counting usually relies on manual field observation and counting, and then prediction is made based on the observation results. The problem with this method is that

it takes a lot of time and human resources and is susceptible to human factors, such as counting errors caused by subjective bias or fatigue of the observer, and the traditional method has been unable to meet the development needs of modern smart agriculture [3]–[5]. With the development of computer vision, the method of counting using deep learning has gradually replaced the traditional manual counting method. Meanwhile, intelligent mobile devices such as tracked vehicles and robot dogs have been widely used in the fields of industrial transportation, logistics handling, and rescue expeditions [6], [7]. The use of intelligent mobile devices such as robot dogs for orchard work is part of the development of smart agriculture [8]. The application of this technology demonstrates the trend of digitalization and automation in the field of agriculture, which helps to improve the efficiency of orchard management and promote the development of agriculture towards intelligence and sustainability. Therefore it is of far-reaching research significance to utilize smart mobile devices to improve the efficiency of orchard counting while improving the counting accuracy in complex orchard scenarios.

Currently, the methods for fruit counting are mainly based on the fruit videos collected by cameras, and the detection algorithms and tracking algorithms of deep learning are utilized for fruit counting [9]–[12].

With the development of deep learning, detection and tracking algorithms are becoming more and more mature, the counting work using detection and tracking methods has become a mainstream method nowadays, there is no expensive equipment cost at the same time, but also real-time detection and tracking counting can be carried out on the limited resources of the equipment. Parico et al. [9] used improved YOLOV4 and DeepSort algorithms for frame-by-frame detection and tracking of fruity pears and finally count the number of fruits. Wang et al. [10] proposed a mango counting method for multi-target tracking, using MangoYOLO to detect the fruit targets in each frame, then Kalman filtering and Hungarian matching algorithm to realize the tracking of fruit targets, and finally count the number of fruits. Gao et al. [11]

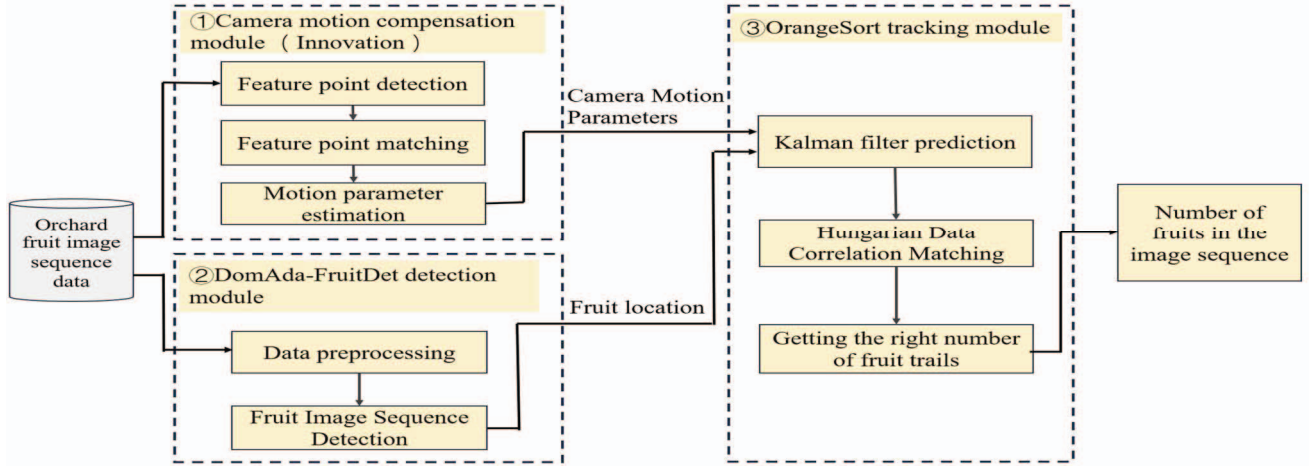


Figure 1. Overall flow chart of counting method

designed YOLOV4-tiny detection algorithm for apple fruit data to determine the fruit location, and then each frame of the fruit location matching to obtain independent fruit ids, and then statistically obtained the number of fruit production. Zhang et al. [12], the head of this research team, designed a YOLOV3-based detection algorithm and an OrangeSort tracking algorithm for citrus fruits used a tracked vehicle to collect fruit data, and finally realized fruit counting with the improved detection algorithm. With the continuous improvement of the detection model and tracking model, the mainstream detection and tracking methods have greatly improved in fruit counting accuracy and speed compared to the traditional methods.

In current fruit counting studies, when utilizing smart mobile devices such as robot dogs to collect data in orchards, the camera lenses often experience vibrating due to walking in complex terrain. Existing mainstream counting methods, namely detection and tracking methods, mostly rely on Kalman filtering algorithms and Hungarian matching algorithms to match and correlate the detected results. Kalman filters are essentially linear gaussian systems and are not suitable for predicting fruit positions in vibrated scenarios. In complex orchard environments, camera movements lead to decreased tracking performance, affecting the accuracy of the final count. Therefore, this paper addresses the issue that existing Kalman filters are not suitable for predicting fruit positions in vibrated scenarios, which affects the performance of the tracker and the accuracy of the final count. This paper proposes a fruit tracking algorithm based on camera motion compensation, named StableSort-CMC (Camera Motion Compensation). This algorithm combines camera motion compensation algorithms with Kalman filters. Traditional image registration techniques are used to obtain camera motion parameters for calibrating and updating the predicted values of the Kalman filter, ultimately achieving fruit counting in vibrated environments.

II. METHOD

The methodology of this paper is mainly divided into two parts. The first part introduces the camera motion compensation algorithm, which innovatively addresses

the issue of inaccurate counting caused by vibrating in actual orchard scenes by combining detection and tracking algorithms. The second part outlines the overall process of fruit counting method. In this part, the input fruit image sequence data is obtained by cutting frames from the collected fruit videos, thus obtaining continuous fruit image data. Through three modules: camera motion compensation module, detection module, and tracking module, the number of fruits contained in the fruit image sequence is obtained. The overall process is illustrated in Fig. 1.

A. Camera Motion Compensation Algorithm

The tracking method based on detection results is essentially a data association process, where the detection results of the previous frame are used to determine whether the fruits in the next frame are associated with those in the previous frame. Fruits that are associated are assigned the same ID to represent the same fruit. For newly appearing fruits, it is first determined whether they have appeared in previous frames. If they have, they are assigned the previous ID to prevent duplicate counting. If not, a new ID is assigned. The output of the data association, i.e., the tracking algorithm, is the result of Hungarian matching. The input matching algorithm includes the predicted position information from the Kalman filter and the detected position information from the next frame. Therefore, to obtain ideal tracking results, the predicted values of the Kalman filter are crucial.

In scenarios without vibration, such as uniform motion, the Kalman filter typically exhibits good predictive ability because it is essentially a linear Gaussian system. In such scenarios, the image sequences captured by the camera only undergo translation. However, in vibrated scenarios, the image sequences also undergo rotation and scaling transformations, which are difficult for the Kalman filter to predict. Therefore, this paper proposes using a camera motion compensation algorithm to calculate the rotation, scaling, and translation transformations between image sequences. These transformations are then incorporated into the Kalman filter to enhance its predictive accuracy.

To compensate for the inaccurate prediction of the Kalman filter in vibrated scenarios, as illustrated in Figure

1 at ①, this paper utilizes traditional image registration [13] methods, including feature point detection, feature point matching, and motion parameter estimation, to obtain camera motion parameter. The camera motion parameter are represented by a 2 rows and 3 columns affine matrix A_{k-1}^k , as shown in (1) [14]:

$$A_{k-1}^k = [M_{2 \times 2} | T_{2 \times 1}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad (1)$$

Here, k and $k-1$ represent adjacent frames, $M_{2 \times 2}$ represents the scaling and rotation transformation between images, and $T_{2 \times 1}$ represents the translation relationship between images. By obtaining these camera motion parameters, the Kalman filter's mean and variance are corrected and updated, thus achieving more stable and accurate tracking counts.

This paper selects image corner points as feature points. Image corner points are typically located at the intersections of two edges or in regions with significant texture changes. They exhibit robustness to changes in illumination and image rotation, making them widely used in motion detection, image matching, video tracking, 3D modeling, and object recognition. In this paper, optical flow is used to match the detected corner points, and an affine transformation function is employed to calculate the transformation relationship of the matched corner points.

In summary, this paper utilizes image registration methods to acquire the rotational, scaling, and translational transformations between images, which are regarded as camera motion parameter. Integrating these transformations into the Kalman filter enhances its predictive capability in vibrated scenarios, thus further improving the performance of the tracker.

B. Overall Method of Counting

The counting process of this paper is shown in Fig. 1, which is mainly divided into three modules camera motion compensation module, detection module and tracking module. First, for the input fruit image sequence, the camera motion compensation module and the detection module calculate the camera motion parameters and detect the fruit position information frame by frame,

respectively; then, the tracking module performs data correlation of the detection results of the neighboring images, and at the same time, the camera motion parameters are used to improve the accuracy of the data correlation of the tracking module, and finally, the number of correctly tracked fruits is obtained.

The camera motion compensation module is shown as ① in Fig. 1, the main method is to detect the feature points frame by frame for the input fruit image sequence, match the detected feature points of two adjacent frames with the sparse optical flow method, and calculate the affine matrix of the image transformation, i.e., the camera motion parameter, with the matched feature points. The obtained camera motion parameters are used to correct and update the mean and variance of the Kalman filter, where the correction and updating process of the mean and variance of the Kalman filter in this paper follows the method in the literature [14], i.e., the camera motion parameters are used to do the corresponding matrix operations on the mean and variance of the Kalman filter. The detection module is shown as ② in Fig. 1, and the detection algorithm DomAda-FruitDet [15] proposed in the team's previous research work is used as the detection algorithm for fruit counting in this paper, which is an Anchor-free domain-adaptive fruit detection algorithm improved based on CenterNet [16], and performs good detection accuracy on a wide range of fruits. The main method is to cut frames from the captured orchard fruit video data to obtain fruit image sequences, then scale normalize the fruit images and feed them into the DomAda-FruitDet detection model to obtain fruit location information.

The tracking module is shown as ③ in Fig. 1. In this paper, we use the OrangeSort [12] algorithm proposed in the early-stage research work of the team as the tracking algorithm. The OrangeSort algorithm is an improved tracking algorithm based on the Sort algorithm [17] and designed using motion displacement estimation method. It can effectively improve the tracking quality, and the specific region counting strategy it employs reduces the problem of repeated counting. It has shown good counting performance on the citrus dataset. At the same

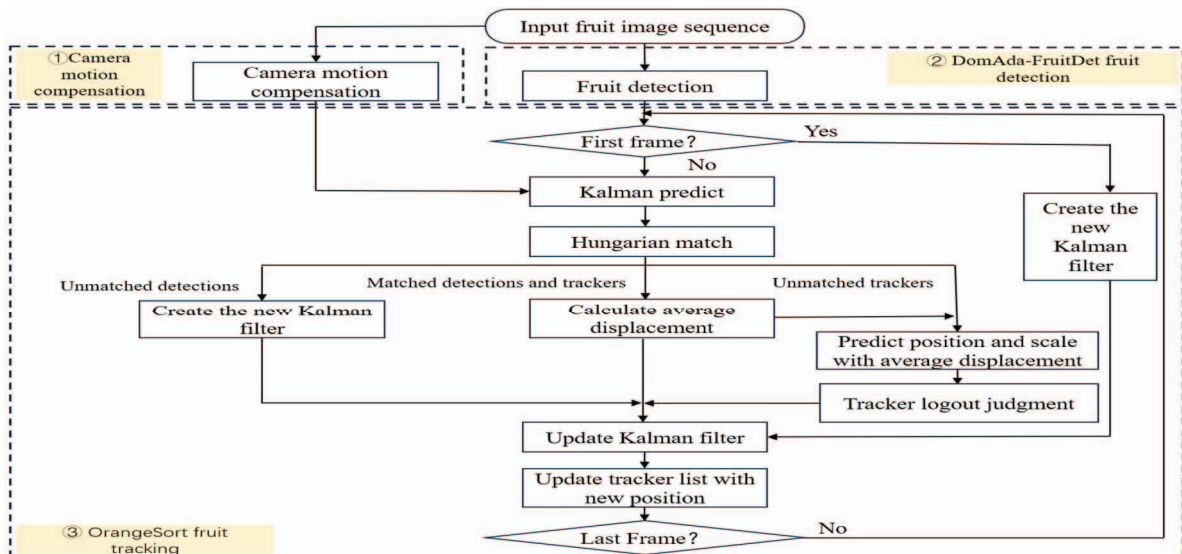


Figure 2. Tracking algorithm StableSort-CMC architecture

time, this paper combines the tracking algorithm with the Camera Motion Compensation (CMC) module to achieve fruit counting in vibrated scenes. The specific process is shown in Fig. 2, and the specific steps are as follows:

First, the detection results and camera motion parameters of the fruit image sequence are obtained. Then, the detection results and camera motion parameters are utilized to calculate the motion matching degree between the Kalman filter predicted tracking results and the detection results of the current frame. Calculate the IoU cost matrix based on the detection boxes and tracking boxes. Finally, the Hungarian matching algorithm will be utilized to associate the detection results with existing tracking results and output three types of results: "unmatched detection targets," "correctly matched detection targets," and "unmatched tracking targets." For the correctly matched detection targets, estimate the position of the unmatched tracking targets in the current frame using the average motion displacement of the tracked fruit targets and then update the Kalman filter state accordingly. Create new Kalman filters for unmatched detection targets. For fruits exceeding the set counting region, delete the corresponding trackers. The counting region strategy in this paper follows the setting in literature [12], which designates the middle 3/5 region of the image as the counting region. For fruits within the counting region, the total number of fruit image sequences is obtained by detecting and tracking frame by frame.

In summary, this paper proposes a counting method that integrates camera motion compensation. By utilizing the obtained camera motion parameters to correct and update the Kalman filter's prediction of target motion states, the performance of the tracker is improved. This further achieves accurate fruit counting in actual orchard scenarios with vibrated scenes.

III. RESULT

A. Data Acquisition

This paper collected the nectarine data on June 16, 2023, at the Beijing Alumni Base of Jilin University in Shunyi District, Beijing. Data was collected in the orchard through the robot dog CyberDog (Xiaomi Inc., Beijing, China) equipped with a gimbal camera, and eleven nectarine videos were collected.

The following acquisition method was followed:

1) The CyberDog traveled among the fruit tree ridges and captured bilateral video sequences of each row of fruit trees.

2) Set the camera to record the video frame rate to ensure that approximately 30 images are captured on each side of each tree, for a total of approximately 60 images on the left and right sides of a tree.

B. Experimental Details

In this paper, a computer equipped with GEFORCE GTX 3090 type GPU and Intel i7 8th model CPU is used for training and testing validation of fruit detection and tracking algorithm. For the detection dataset, a total of 678 nectarine images were produced and randomly disrupted to divide the training and validation sets by 7:3. For the tracking dataset, a total of 1061 nectarine image sequences were produced.

To verify the effectiveness of the counting method proposed in this paper, we primarily utilize the counting evaluation metrics from literature [12], comparing the counting errors derived from manual counting and algorithmic counting to assess the counting performance. The evaluation metrics also include accurate counting, erroneous counting (mistaking the leaf background as tracking targets for counting), double counting (counting the same fruit target multiple times), and missed counting (omitting tracked fruit targets).

To validate the effectiveness of the tracking algorithm presented in this paper, the evaluation metrics proposed in the MOT (Multiple Object Tracking)[18].challenge were used as the standard for assessment. In the evaluation of multi-object tracking algorithms, the primary focus is on the MOTA (Multiple Object Tracking Accuracy) and MOTP (Multiple Object Tracking Precision) metrics. The MOTA mainly concerns the number of errors made by the tracker, with fewer errors leading to a higher MOTA score, indicating better tracker performance. Its formula is given by:

$$MOTA = 1 - \frac{\sum(FN+FP+IDS)}{\sum GT} \quad (2)$$

where FN represents the number of times the tracker failed to detect a target, FP represents the number of times the tracker falsely predicted a target, IDS represents the number of times an identity switch occurred, and GT represents the total number of actual targets.

C. Experimental Results and Analysis

1) Comparative Experiment

In this section's experiment, to demonstrate the effectiveness of the proposed multi-object tracking algorithm StableSort-CMC, two tracking algorithms, ByteTrack [19] and Bot-Sort [14] were chosen for comparison. ByteTrack is an algorithm that relies solely on the Kalman filter and Hungarian matching algorithm for tracking, similar to other non-ReID algorithms, using only the bounding box (bbox) obtained from object tracking. The feature of the ByteTrack algorithm is its use of low-confidence boxes, performing a secondary matching for the tracking algorithm, effectively optimizing the tracking process and addressing ID switching issues due to occlusions. Bot-Sort is a new robust tracking algorithm that combines the advantages of camera motion and image appearance information, introducing camera motion compensation into the tracking algorithm for the first time and setting a more accurate Kalman filter state vector.

The comparison experiment utilized the DomAda-FruitDet algorithm for detection, with a detection algorithm accuracy metric mAP_{0.5} of 0.783. The counting comparison results are shown in Table 1, where our algorithm counted 132 nectarines, closest to the manual count of 84, with the lowest counting error of only 0.571. Next, the Bot-Sort algorithm counted 138, with a counting error of 0.642.

TABLE I. COMPARISON OF COUNTING ACCURACY OF DIFFERENT TRACKING ALGORITHMS

Tracking algorithm	Manual counting	Number of fruit counting	Number of correct counts	Number of false counts	Number of double-counts	Counts error
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					counting instances	
ByteTrack	84	143	72	36	35	0.702
Bot-Sort	84	138	77	30	31	0.642
StableSort-CMC	84	132	75	33	24	0.571

The tracking comparison results are presented in Table 2, where our algorithm achieved the highest scores in MOTA and MOTP accuracy, at 63.4% and 74.6%, respectively. Following was the Bot-Sort algorithm, with MOTA and MOTP at 59.4% and 73.2%. Thus, in vibrated scenarios, tracking algorithms that incorporate a camera motion compensation module are more competitive in counting accuracy and tracking effectiveness than those relying solely on the Kalman filter. Moreover, among algorithms that also incorporate camera motion compensation, the algorithm proposed in this paper outperforms Bot-Sort in terms of tracking methods and strategies.

TABLE II. COMPARISON OF TRACKING ACCURACY OF DIFFERENT TRACKING ALGORITHMS

Tracking algorithm	MOTA	MOTP	FP	FN	IDS
ByteTrack	55.5	72.4	131	936	35
Bot-Sort	59.4	73.2	104	871	31
StableSort-CMC	63.4	74.6	178	706	24

In summary, the proposed algorithm, StableSort-CMC, has achieved competitive results on nectarine data collected by robotic dogs, effectively improving the tracking performance and final counting accuracy of the tracking algorithm.

2) Ablation Study

To verify the effectiveness of the camera motion compensation module in enhancing tracker performance and counting accuracy in vibrated scenarios, two ablation

experiments were conducted on the tracking and counting aspects of our algorithm, both based on the DomAda-FruitDet detection algorithm with a detection accuracy of mAP 0.5 at 0.783.

For tracking performance, Table 3 shows the change in tracking accuracy before and after incorporating the camera motion compensation module (CMC) into the OrangeSort algorithm. Adding the CMC module to the OrangeSort tracking algorithm improved MOTA accuracy by 2.9%, and reduced the number of false positives (FP), false negatives (FN), and ID switches (IDS). This indicates that calibrating and updating the Kalman filter with parameters corrected for camera motion can more accurately predict the position of nectarines, thereby enhancing the tracker's performance.

TABLE III. CAMERA MOTION COMPENSATION MODULE TRACKING ABLATION EXPERIMENT

Tracking algorithm	MOTA	FP	FN	IDS
OrangeSort	60.5	196	755	32
OrangeSort+CMC	63.4	178	706	24

TABLE IV. CAMERA MOTION COMPENSATION MODULE COUNTING ABLATION EXPERIMENT

Tracking algorithm	Manual counting	Number of fruit counting	Number of correct counts	Number of false counts	Number of double-counting instances	Counts error
OrangeSort	84	140	74	34	32	0.667
OrangeSort+CMC	84	132	75	33	24	0.571

For counting accuracy, Table 4 illustrates the change in counting accuracy before and after adding the camera motion compensation module to the OrangeSort algorithm. After introducing the camera motion



Figure 3. Robot dog and visualization of tracking results before and after adding camera motion compensation. (a) shows the Xiaomi mechanical dog CyberDog, the data acquisition device in this paper. (b.1), (b.2), (c.1), (c.2) The two yellow lines in the four nectarine plots are divided counting areas, which ideally should always have the same number of ids for the same nectarine. (b.1), (b.2) shows that the two nectarines with id 5 and 6 on the right side of the image in 5th frame before adding the camera motion compensation, after masking, reappeared in 8th frame obtaining new id 8 and 7, respectively, and repeated counts were performed for each of the two nectarines. (c.1) and (c.2) demonstrates that the corresponding two nectarines always have the same id after adding camera motion compensation.

compensation module, the counting error decreased by 9.6%, with an increase in accurate counts and a decrease in erroneous and duplicate counts. This demonstrates that the OrangeSort tracking algorithm, when combined with the camera motion compensation module, effectively improves counting accuracy in vibrated scenarios.

Fig. 3 shows the tracking results and the robotic dog collection device. The results with camera motion compensation show a reduction in duplicate counting. In the images from the algorithm with the camera motion compensation module, the ID of two nectarines are consistently maintained as 4 and 5, whereas, in the images from the algorithm without the camera motion compensation module, the IDs of the same two nectarines change from 5 and 6 to 8 and 7, indicating that the same nectarines were counted twice.

IV. CONCLUSION

Building upon previous research, this paper proposes a tracking algorithm called StableSort-CMC, combined with detection algorithm called DomAda-FruitDet, to address the issue of inaccurate fruit counting in vibrated scenarios due to the performance degradation of tracking algorithms. The StableSort-CMC algorithm utilizes traditional image registration methods to obtain camera motion parameters, which are then used to correct and update the Kalman filter with the rotational, scaling, and translational relationships between image sequences. Through ablation experiments, the effectiveness of the camera motion compensation module is demonstrated, and comparative experiments confirm that the proposed tracking algorithm effectively improves the performance of the tracker in vibrated scenarios, thereby enhancing the accuracy of fruit counting in orchards under vibrated conditions.

In the future, we aim to design lightweight model architectures for deployment on edge devices. Additionally, we plan to integrate these edge devices onto mobile robot dogs to achieve real-time fruit counting in orchards. To mitigate the impact of vibrated scenarios on the tracker, we intend to utilize image registration methods as an alternative to tracking algorithms, thus avoiding frame-by-frame detection and tracking and further reducing counting errors caused by vibration.

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REFERENCES

- [1] H. Tian, T. Wang, Y. Liu, X. Qiao, and Y. Li, "Computer vision technology in agricultural automation—A review," *Information Processing in Agriculture*, vol. 7, no. 1, pp. 1-19, 2020.
- [2] Y. Feng, W. Ma, Y. Tan, H. Yan, J. Qian, Z. Tian, and A. Gao, "Approach of Dynamic Tracking and Counting for Obscured Citrus in Smart Orchard Based on Machine Vision," *Applied Sciences*, vol. 14, no. 3, pp. 1136, 2024.
- [3] R. Wang, R. Li, T. Chen, J. Zhang, C. Xie, K. Qiu, P. Chen, J. Du, H. Chen, and F. Shao, "An automatic system for pest recognition and forecasting," *Pest Management Science*, vol. 78, no. 2, pp. 711-721, 2022.
- [4] J. Torres-Sánchez, J. Souza, S. F. di Gennaro, and F. J. Mesas-Carrascosa, "Fruit detection and yield prediction on woody crops using data from unmanned aerial vehicles," *Frontiers in Plant Science*, vol. 13, pp. 1112445, 2022.
- [5] C. Poudyal, L. F. Costa, H. Sandhu, Y. Ampatzidis, D. C. Otero, O. C. Arbelo, and R. H. Cherry, "Sugarcane yield prediction and genotype selection using unmanned aerial vehicle-based hyperspectral imaging and machine learning," *Agronomy Journal*, vol. 114, no. 4, pp. 2320-2333, 2022.
- [6] D. Oladimeji, K. Gupta, N. A. Kose, K. Gundogan, L. Ge, and F. Liang, "Smart transportation: an overview of technologies and applications," *Sensors*, vol. 23, no. 8, pp. 3880, 2023.
- [7] Y. Song, F. R. Yu, L. Zhou, X. Yang, and Z. He, "Applications of the Internet of Things (IoT) in smart logistics: A comprehensive survey," *IEEE Internet of Things Journal*, vol. 8, no. 6, pp. 4250-4274, 2020.
- [8] M. Bergerman, S. M. Maeta, J. Zhang, G. M. Freitas, B. Hamner, S. Singh, and G. Kantor, "Robot farmers: Autonomous orchard vehicles help tree fruit production," *IEEE Robotics & Automation Magazine*, vol. 22, no. 1, pp. 54-63, 2015.
- [9] A. I. B. Parico, and T. Ahamed, "Real time pear fruit detection and counting using YOLOv4 models and deep SORT," *Sensors*, vol. 21, no. 14, pp. 4803, 2021.
- [10] Z. Wang, K. Walsh, and A. Koirala, "Mango fruit load estimation using a video based MangoYOLO—Kalman filter—hungarian algorithm method," *Sensors*, vol. 19, no. 12, pp. 2742, 2019.
- [11] F. Gao, W. Fang, X. Sun, Z. Wu, G. Zhao, G. Li, R. Li, L. Fu, and Q. Zhang, "A novel apple fruit detection and counting methodology based on deep learning and trunk tracking in modern orchard," *Computers and Electronics in Agriculture*, vol. 197, pp. 107000, 2022.
- [12] W. Zhang, J. Wang, Y. Liu, K. Chen, H. Li, Y. Duan, W. Wu, Y. Shi, and W. Guo, "Deep-learning-based in-field citrus fruit detection and tracking," *Horticulture Research*, vol. 9, 2022.
- [13] M. Deshmukh, and U. Bhosle, "A survey of image registration," *International Journal of Image Processing (IJIP)*, vol. 5, no. 3, pp. 245, 2011.
- [14] N. Aharon, R. Orfaig, and B. Bobrovsky, "BoT-SORT: Robust associations multi-pedestrian tracking. arXiv 2022," arXiv preprint arXiv:2206.14651.
- [15] W. Zhang, C. Zheng, C. Wang, and W. Guo, "DomAda-FruitDet: Domain-Adaptive Anchor-Free Fruit Detection Model for Auto Labeling," *Plant Phenomics*, vol. 6, pp. 0135, 2024.
- [16] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "Centernet: Keypoint triplets for object detection." pp. 6569-6578.
- [17] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking." pp. 3464-3468.
- [18] P. Dendorfer, H. Rezaatofighi, A. Milan, J. Shi, D. Cremers, I. Reid, S. Roth, K. Schindler, and L. Leal-Taixé, "Mot20: A benchmark for multi object tracking in crowded scenes," arXiv preprint arXiv:2003.09003, 2020.
- [19] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "Bytetrack: Multi-object tracking by associating every detection box." pp. 1-21